



INFECTION RATE FOR ALCOHOLISM IN MEXICO

THE ALCOHOLISM MODELED AS A CONTAGIOUS DISEASE

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Summary

There are interesting articles that consider alcoholism as an epidemic disease, and there are some survey about Alcoholism. With this data we will get β (Infection rate) and α (recuperation rate) of a SIR Model Equation for an epidemic model. The equilibrium points, the maximum infection rate and when it will happen are going to be determined in this project.

In chapter one using Euler method and SIR equation for epidemic disease we will first approximate $\beta > 0$ with $\alpha = 0$, then $\beta = 0$ and $\alpha > 0$ and at last $\beta > 0$ and $\alpha > 0$ to look like a epidemic disease with some equilibrium point in the distance. In chapter two we are going to validate the equilibrium point that we got with some analytical method .

For Mexican's data we obtained $\alpha = 0.0014$ and $\beta = 0.06$ and the equilibrium point ($S_{\infty} = 12.25, I_{\infty} = 0, R_{\infty} = 87.75$) and the maximum infection rate in $I(37.75) = 23.04$ for every 100 people. These are unstable points, but realistically we only could get positive eigenvalues .

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Chapter 1

Introduction

In 2012 the Mexican Commission against Addictions published its 2011 national survey on both addictions in general and on alcohol in particular. This work has been developed on a regular basis given the government's concern about the increase in consumption over the past few years, considering alcoholism as a public health problem . This increase has been explained due to the perception of risk that has decreased while the social tolerance has increased.[3].

On the other hand there are interesting articles that consider alcoholism as an epidemic disease, one of them was published by Brandy Benedict in 2007, who fashions to alcoholism as an epidemic disease among alcohol consumers.[1].

Since mathematical model can mimic the process of alcoholism and provide useful methods to control the spread of drinking behavior the aim of this study is to use the SIR model described by Diego de Pereda Sebastian [4] and Mexican Surveys to determine the infection rate that exists acquiring the disease when exposed between consumers of alcohol.

It is difficult to understand the meaning of R function in this case, but we will use it to describe the people that were totally rehabilitated in a rehabillitation program or were exposed to intense social programs of alcoholism prevention.

Chapter 2

Euler's Method Calculation of α and β

Alcohol abuse is one of the public health issues of our time, this health phenomenon affects to all people as same , regardless of gender, age, social stratum or region of our countries.

Epidemiological studies, both national and worldwide, show that this phenomenon is changing and has been increasing over time.

For these reasons in Mexico, epidemiological studies have been developed with a periodicity of approximately six years. The last one was published in 2012, last year a survey was conducted and the results are going to be released in 2017.

In the epidemiological studies of 2012, the table 2.1 was published [3]. We will use it to take the numbers in order to work with our equation system for an epidemic model.

Gender/Year	2002	2008	2011.
b. All Dependence	4.1	5.0	6.2
d. Men Dependence	8.3	8.5	10.8
f. Woman Dependence	0.6	1.8	1.8

Table 2.1: $I(t)$. Percentage of the Population with Alcohol Dependence [3]

For now we will take dependence like alcoholic infection and only the years 2008 ($t = 0$) and 2011 ($t = 3$). Then 2008 will be $t = 0$

The equation system for an epidemic model is showed in equation 2.1 :

Cuadro 1: Tendencias del Consumo de Alcohol. Población Total de 12 a 65 años				
	Encuestas Nacionales de Adicciones			
	2002	2008	2011	
Total		%	%	IC 95%
Consumo Alguna Vez	64.9	61.3	71.3	70.071-72.609
Consumo Último Año	46.3	44.1	51.4	50.017-52.731
Consumo Último mes	19.2	27.3	31.6	30.355-32.799
Consumo alto	NC	32.0	32.8	31.498-34.108
Consumo diario	1.4 ⁴	.9	.8	0.590-0.964
Consuetudinarios	5.6	6.1	5.4	4.806-5.956
Dependencia	4.1	5.0	6.2	5.553-6.754
Hombres				
Consumo Alguna Vez	78.6	72.3	80.6	79.286-81.990
Consumo Último Año	61.1	55.9	62.7	60.836-64.538
Consumo Último mes	33.6	38.8	44.3	42.445-46.094
Consumo alto	NC	45.0	47.2	45.348-49.070
Consumo diario	2.5	1.6	1.4	1.055-1.792
Consuetudinarios	9.7	10.5	9.6	8.539-10.694
Dependencia	8.3	8.5	10.8	9.672-11.981
Mujeres				
Consumo Alguna Vez	53.6	51.0	62.6	60.804-64.458
Consumo Último Año	34.2	33.2	40.8	38.959-42.598
Consumo Último mes	7.4	16.6	19.7	18.378-20.999
Consumo alto	NC	19.9	19.3	17.876-20.743
Consumo diario	.4	.3	.2	0.057-0.287
Consuetudinarios	2.3	2.1	1.4	1.066-1.762
Dependencia	.6	1.8	1.8	1.368-2.187

Fuente: Encuestas Nacionales de Adicciones 2002, 2008 y 2011.

Figure 2.1: **Results:** Cuadro de Encuestas Nacional de Adicciones 2002, 2008 y 2012 [3]

$$\left\{ \begin{array}{l} \frac{dS}{dt} = -\beta S(t)I(t), \quad P(0) = 95 \wedge P(3) = 93.8, \\ \frac{dI}{dt} = \beta S(t)I(t) - \alpha I(t), \quad I(0) = 5 \wedge I(3) = 6.2, \\ \frac{dR}{dt} = \alpha I(t) \quad R(0) = 0. \end{array} \right. \quad (2.1)$$

Where:

$$\begin{aligned} S(t) + I(t) + R(t) &= 100 \quad \forall t, \\ t &= \text{Years from 2008} \\ S(t) &= \text{Susceptible population for every 100 individuals} \\ I(t) &= \text{Population with dependence problem for every 100 individuals} \\ R(t) &= \text{Totally Rehabilitated Population for every 100 individuals} \\ \beta &= \text{Infection rate} \\ \alpha &= \text{Recuperation rate} \\ \frac{1}{\alpha} &= \text{Time of recuperation from alcoholic problem in years} \end{aligned}$$

(2.2)

2.1 Calculating $\beta > 0$ with $\alpha = 0$ from a SIR model

With Python and Euler Method we will approximate the β value to get $I(3)$.

The equation system for an epidemic model with some numbers and $\alpha = 0$.

$$\begin{cases} \frac{dS}{dt} = -\beta S(t)I(t), & P(0) = 95 \wedge P(3) = 93.8 \\ \frac{dI}{dt} = \beta S(t)I(t), & I(0) = 5 \wedge I(3) = 6.2 \end{cases} \quad (2.3)$$

First thing is to determine the appropriate size of Δt (stepsize) in such a way that we have an error smaller than 0.01 in $t = 5$. Provisionally we set numbers to $\beta = 0.001$ and $\alpha = 0$.

The arithmetical values that we got are in the Table 2.2.

Δt	$S(5)$	$I(5)$	$S(5)_t - S(5)_{t-1}$	$I(5)_t - I(5)_{t-1}$
0.250	92.0548	7.9452	NA	NA
0.125	92.0352	7.9645	-0.0196	0.0196
0.0625	92.0254	7.9746	-0.0098	0.0098

Table 2.2: $\Delta(\beta > 0, \alpha = 0)$. Simulation with Euler to determine Δ

Observe in Table 2.2 that the better valor for Δ is $\frac{1}{16}$.

With $\Delta = \frac{1}{16}$ we obtain that $\beta \approx 0.00076$. See the figure 2.2.

2.2 Calculating $\alpha > 0$ with $\beta = 0$ from a SIR model

With Python and Euler Method we approximate the α value to get $I(3) = 6.2$.

The equation system for an epidemic model with some values and $\beta = 0$.

$$\begin{cases} \frac{dI}{dt} = -\alpha I(t), & I(0) = 5 \wedge I(3) = 6.2, \\ \frac{dR}{dt} = \alpha I(t). & R(0) = 0 \end{cases} \quad (2.4)$$

The equation (2.6) look like (2.3) with $I(t) = S(t) * I(t)$ and $\alpha = \beta$.

First we determine the appropriate size of Δt (stepsize) in such a way that we have an error smaller than 0.01 in $t = 5$. Because $\frac{1}{\alpha}$ is the recuperation time, we set arithmetical values to $\beta = 0$ and $\alpha = 0.10$. Guessing that someone with alcoholic problems will take

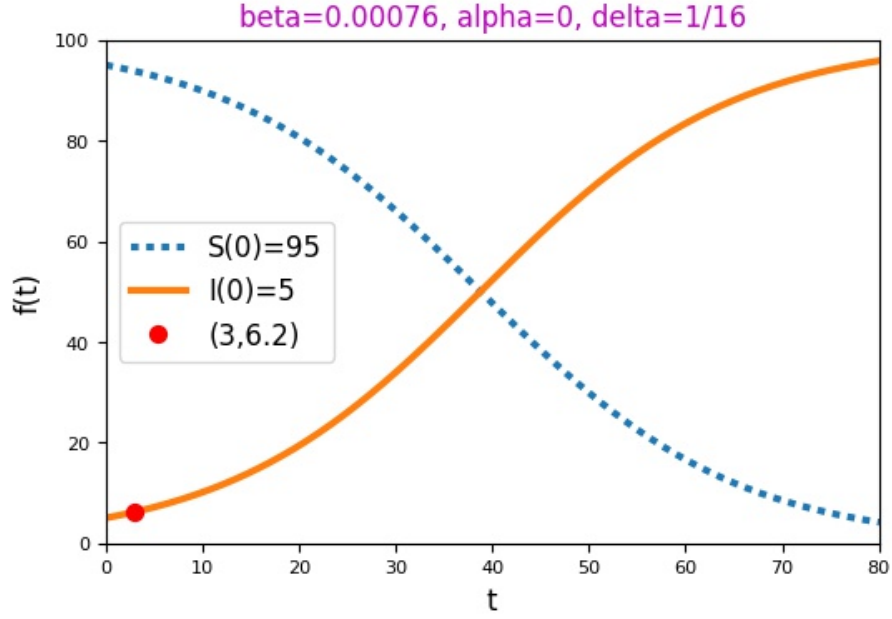


Figure 2.2: $\Delta = \frac{1}{16}, \beta > 0, \alpha = 0$. Euler's Simulation to determine β .

10 years to rehabilitate. For this reason we always choose $\alpha < 1$ because it will take at least 1 year to rehabilitate someone.

The arithmetical values that we get are in the Table 2.3 :

Δt	$I(5)$	$R(5)$	$I(5)_t - I(5)_{t-1}$	$R(5)_t - R(5)_{t-1}$
0.125	8.2181	1.7119	NA	NA .
0.0625	8.2308	1.7692	0.0127	-0.0127
0.03125	8.2372	1.2628	0.0064	-0.0064

Table 2.3: $\Delta(\alpha > 0, \beta = 0)$. Euler's Simulation to determine what value of Δ

Observe in the table 2.3 that the better valor for Δ is $\frac{1}{32}$

You can see in figure 2.3 that with $\Delta = \frac{1}{32}$, $I(0) = 5$ and $R(0) = 0$ the behavior with $\alpha = 0.10$.

2.3 Calculating $\alpha > 0$ and $\beta > 0$ from a SIR model

The Equation System is the same to the equation 2.1.

Because $\frac{1}{32} < \frac{1}{16}$, we choose $\Delta = \frac{1}{32}$ to simulate values for β and α and to obtain $I(3) = 6.2$. We have many solutions, all of them have the same proportions $R_0 = \frac{\beta}{\alpha} < 1$. This means

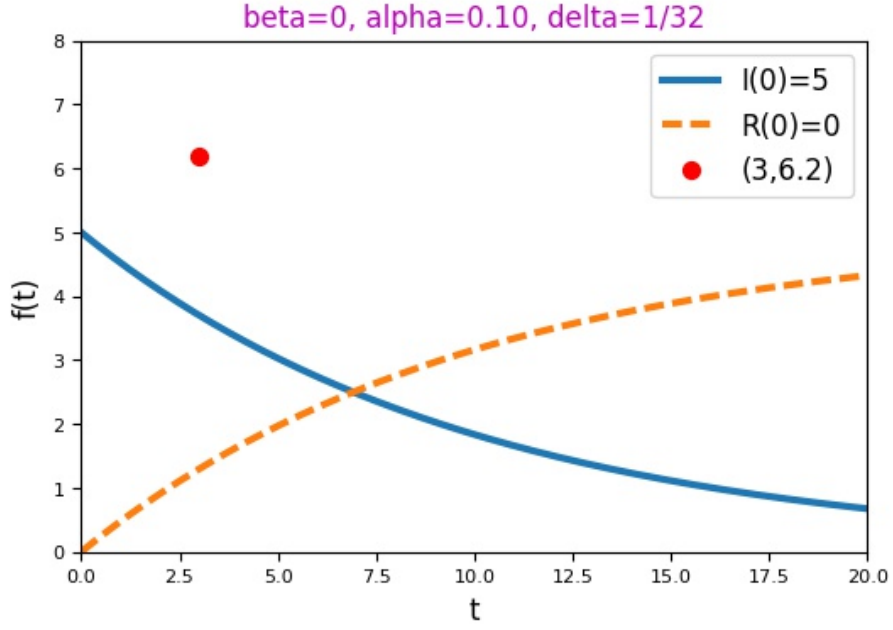


Figure 2.3: $\Delta = \frac{1}{32}, \alpha > 0, \beta = 0$. Euler's Simulation to determine α .

(Diego de Pereda [4]) that the infection disease will increase and eventually it will decrease $I(t) = 0$.

One of them is $\beta \approx 0.0014$ and $\alpha \approx 0.06$. This was chosen because the recuperation rate is $\frac{1}{0.06} = 16.67$.

2.4 Equilibrium points

It is known that equilibrium point is obtained where $\frac{df(t)}{dt} = 0$, and in this case when:

$$\left\{ \begin{array}{lcl} \frac{dS}{dt} & = & -0.0014S(t)I(t) = 0, \\ \frac{dI}{dt} & = & 0.0014S(t)I(t) - 0.06I(t) = 0, \\ \frac{dR}{dt} & = & 0.06I(t) = 0. \end{array} \right. \quad (2.5)$$

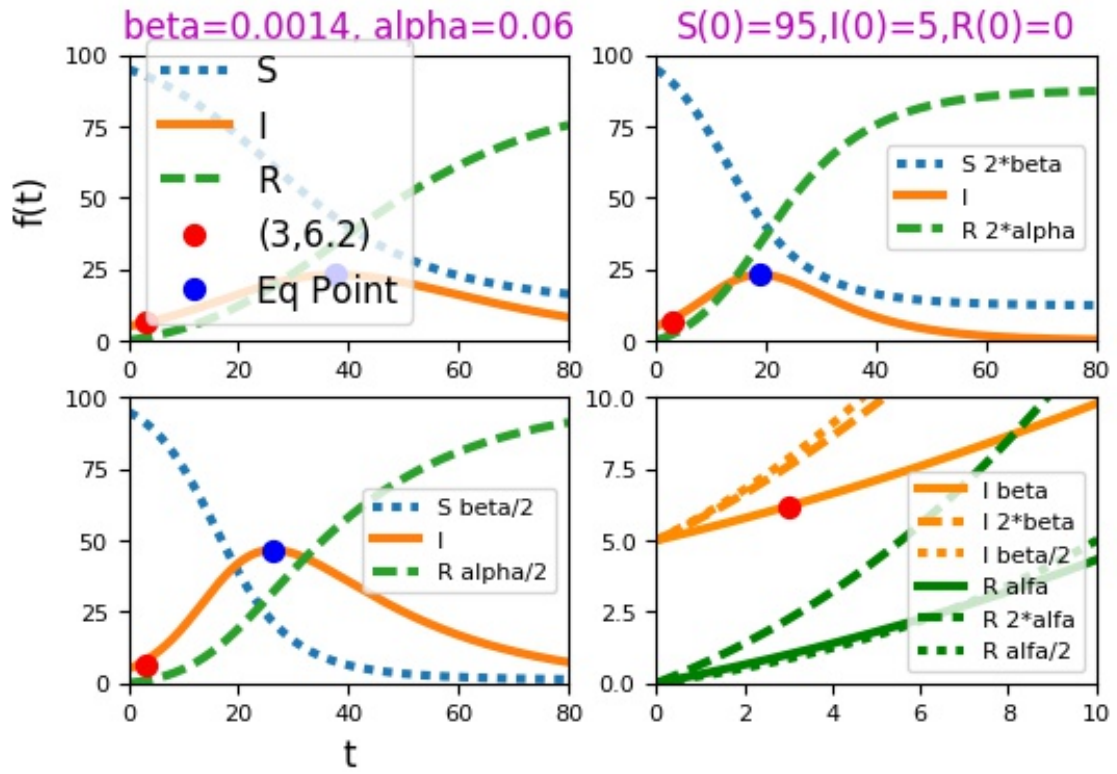


Figure 2.4: $\Delta = \frac{1}{32}, \alpha > 0, \beta > 0$. Euler's Simulation with different β and α .

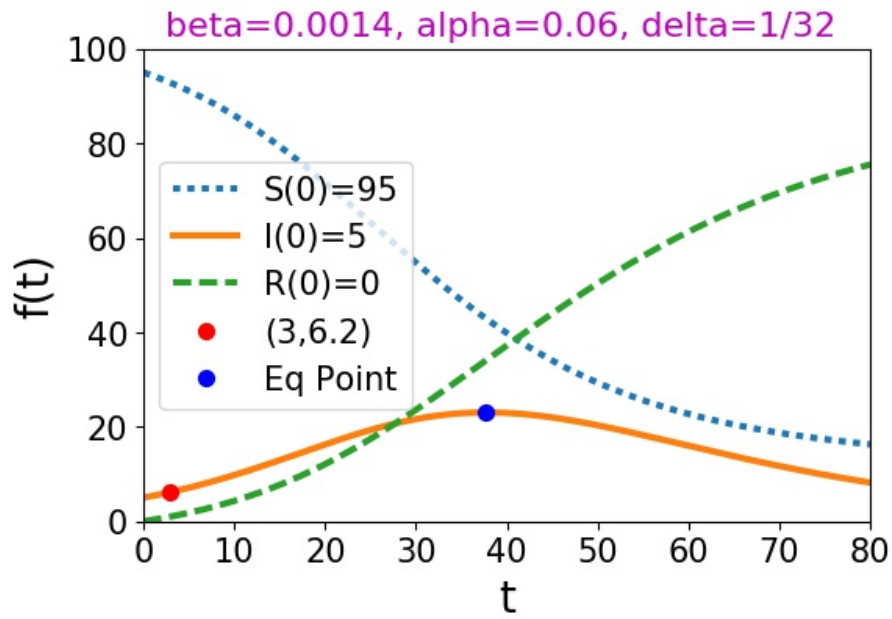


Figure 2.5: $\Delta = \frac{1}{32}, \alpha = 0.0014, \beta = 0.06$. Euler's Simulation to determine β and α .

It is easily perceived that it will happen with $I(t) = 0$. In figure 2.5 we observe that it will be when $t \rightarrow \infty$. The Equilibrium points obtained were simulated with Python in year 1000 and here they are:

$$\begin{aligned} \lim_{t \rightarrow \infty} S(t) &= 12.25, \\ \lim_{t \rightarrow \infty} I(t) &= 0.0, \\ I_{max} &= 23.04 \quad t = 37.75, \\ \lim_{t \rightarrow \infty} R(t) &= 87.75. \end{aligned} \tag{2.6}$$

Chapter 3

Analytic Validation

In the previous chapter we calculated with Euler's Method the arithmetical values for α and β to get the point $I(3) = 6.2$. We also calculated the equilibrium point where $\frac{df(t)}{dt} = 0$. We will now try to verify this values with an analytic method.

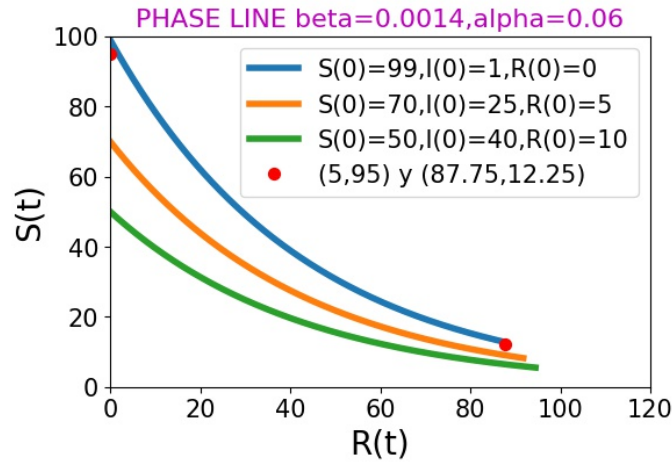


Figure 3.1: **Graph $S(t)$ vs $R(t)$, $I_\infty = 0$** . Euler's Simulation to determine equilibrium point.

In figure 3.1 we simulated different initial values for our model, and found that they get slightly different equilibriums points for everyone.

Diego de Pereda Sebastian [4] established that the I_{max} will be determined for the next equation:

$$I_{max} = S_0 + I_0 - \frac{\alpha}{\beta} \log S_0 - \frac{\alpha}{\beta} + \frac{\alpha}{\beta} \log \frac{\alpha}{\beta}$$

We replace the values and obtained

$$I_{max} = 95 + 5 - \frac{0.0014}{0.06} \log 95 - \frac{0.0014}{0.06} + \frac{0.0014}{0.06} \log \frac{0.0014}{0.06} = 23.03$$

That is similar at $I_{max} = 23.04$ that we previously got.

We now have to found The Jacobine Matrix to calculate the eigenvalues and determine what kind of point we have.

$$\begin{bmatrix} \frac{\partial f_1}{\partial S} & \frac{\partial f_1}{\partial I} & \frac{\partial f_1}{\partial R} \\ \frac{\partial f_2}{\partial S} & \frac{\partial f_2}{\partial I} & \frac{\partial f_2}{\partial R} \\ \frac{\partial f_3}{\partial S} & \frac{\partial f_3}{\partial I} & \frac{\partial f_3}{\partial R} \end{bmatrix}_{(S_0, I_0, R_0)} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix}$$

The Jacobine Matrix equation for SIR Model (2.1) in point (12.25,0,87.75) with $\beta = 0.0014$ and $\alpha = 0.06$ is:

$$\begin{bmatrix} -\beta I & -\beta S & 0 \\ \beta I & \beta S & -\alpha \\ 0 & 0 & \alpha \end{bmatrix}_{(12.25, 0, 87.75)} = \begin{bmatrix} 0 & -0.0175 & 0 \\ 0 & 0.0175 & -0.06 \\ 0 & 0 & 0.06 \end{bmatrix} = \begin{bmatrix} 0.06 \\ 0.0175 \\ 0 \end{bmatrix}$$

Norman Lebovitz [5] established the next important theorem for The Linear Stability with Jacobian Matrix with arbitrary dimension (nxn)

Theorem 8.2.3 *The origin is asymptotically stable for the system if the real part of every eigenvalue is negative. It is unstable if any eigenvalue has a positive real part.*

Therefore it is known that the point is unstable.

But we know that $S > 0$, $I > 0$, $\alpha > 0$, $\beta > 0$ and $I_\infty = 0$ therefore we will always get eigenvalues grater than 0.

Chapter 4

Conclusions

Until now we have obtained some values for $\alpha = 0.06$ and $\beta = 0.0014$, but it seems that they are not logical because it looks like everybody have been rehabilitated and that the infection rate will eventually be 0.

The factors obtained were calculated based on information from 2 years ago, but it would be interesting to make new calculations with the 3 years that we have or the 4 that will be published this year.

this model we have achieved that $I_{\infty} = 0$ but this is not realistic number, since it is a problem that at least in the short term it will not be possible to eradicate. It would then be desirable to obtain an equilibrium point $I_{\infty} > 0$. It is necessary to explore other methods that are more suitable for this purpose

Bibliography

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<http://people.cs.uchicago.edu/~lebovitz/Eodesbook/stabeq.pdf> *Consulted by Jul 2017*

Appendix A

Euler's Method Python Program

```
# Program      : Euler's method for a system
# Author       : MOOC team Mathematical Modelling Basics
# Created      : May, 2017
# Modifie      : June, 2017 by Edith Reyes loya

import numpy as np
import matplotlib.pyplot as plt

print("Solution for dS/dt, dI/dt and dR/dt")

# Initializations
def myeuler(Dt,S_init,I_init,R_init,beta,alfa):
    #Dt = 1/8                                # timestep Delta t
    #P_init = 20                             # initial population of P
    #G_init = 5                               # initial population of G
    t_init = 0                               # initial time
    t_end = 1000                             # stopping time

    n_steps = int(round((t_end-t_init)/Dt)) # total number of timesteps

    X = np.zeros(3)                          # create space for current X=[P,G]^T
    dXdt = np.zeros(3)                      # create space for current derivative
    t_arr = np.zeros(n_steps + 1)           # create a storage array for t
    X_arr = np.zeros((3,n_steps+1))         # create a storage array for X=[P,G]^T
    t_arr[0] = t_init                       # add the initial t to the storage array
    X_arr[0,0] = S_init                     # add the initial S to the storage array
    X_arr[1,0] = I_init                     # add the initial I to the storage array
    X_arr[2,0] = R_init                     # add the initial R to the storage array

    # Euler's method

    for i in range (1, n_steps + 1):
        t = t_arr[i-1]                      # load the time
```



```

        S = X_arr[0,i-1]                # load the value of S
        I = X_arr[1,i-1]                # load the value of I
        R = X_arr[2,i-1]                # load the value of R
        X[0] = S                        # fill current state vector X=[S,I,R]^T
        X[1] = I
        X[2] = R
        dSdt = -beta*S*I
        dIdt = beta*S*I-alfa*I
        dRdt = alfa*I
        dXdt[0] = dSdt                  # fill derivative vector dX/dt
        dXdt[1] = dIdt
        dXdt[2] = dRdt
        Xnew = X + Dt*dXdt              # calculate X on next time step
        X_arr[:,i] = Xnew               # store Xnew
        t_arr[i] = t + Dt              # store new t-value
    return(t_arr,X_arr)

# Plot the results

fig = plt.figure()
delta_inv=32
delta=1/delta_inv
al=0
be=(np.log(6.2)-np.log(5))/3/100
val_t,res0=myeuler(delta,95,5,0,0.0014,0.06)
print(res0[0,3*delta_inv])
print(res0[1,3*delta_inv])
print(res0[2,3*delta_inv])
print(res0[1,3*delta_inv]+res0[0,3*delta_inv]+res0[2,3*delta_inv])
ord=sorted(res0[1,:],reverse=True)
a=np.where(res0[1,:] == ord[0])
print(val_t[a],res0[1,a])
print(res0[0,32000],res0[1,32000],res0[2,32000])
prop=0.06/0.0014
max=100-prop*np.log(95)-prop+prop*np.log(prop)
print(max)
val_t,res1=myeuler(delta,95,5,0,0.0028,0.12)
ord=sorted(res1[1,:],reverse=True)
b=np.where(res1[1,:] == ord[0])
print(val_t[b],res0[1,b])
val_t,res2=myeuler(delta,95,5,0,0.0021,0.045)
ord=sorted(res2[1,:],reverse=True)
c=np.where(res2[1,:] == ord[0])
print(val_t[c],res0[1,c])

#plt.subplot(2,2,1)
plt.plot(val_t,res0[0,:],linewidth = 3,linestyle=":", label="S") # plot P vs. time
plt.plot(val_t,res0[1,:],linewidth = 3,linestyle="-", label="I") # plot P vs. time
plt.plot(val_t,res0[2,:],linewidth = 3,linestyle="--", label="R") # plot P vs. time

```

```

plt.plot(3,6.2,"o",markersize=np.sqrt(50.),color="r", label="(3,6.2)") # plot P vs. time
plt.plot(val_t[a],res0[1,a],"o",markersize=np.sqrt(50.),color="b", label="Eq Point") # p
plt.title('beta=0.0014, alpha=0.06', fontsize = 12,color='m') # set title
plt.ylabel('f(t)', fontsize = 12) # name of vertical axis
plt.xticks(fontsize = 8) # adjust the fontsize
plt.yticks(fontsize = 8) # adjust the fontsize
plt.axis([0,80, 0,100]) # set the range of the axes
plt.legend(fontsize=12) # show the legend

#plt.subplot(2,2,4)
plt.plot(val_t,res0[1,:],linewidth = 3,linestyle="-",color="darkorange",label="I beta")
plt.plot(val_t,res1[1,:],linewidth = 3,linestyle="--",color="darkorange",label="I 2*beta")
plt.plot(val_t,res2[1,:],linewidth = 3,linestyle=":",color="darkorange",label="I beta/2")
plt.plot(val_t,res0[2,:],linewidth = 3,linestyle="-",color="g",label="R alfa") # plot P
plt.plot(val_t,res1[2,:],linewidth = 3,linestyle="--",color="g",label="R 2*alfa") # plot
plt.plot(val_t,res2[2,:],linewidth = 3,linestyle=":",color="g",label="R alfa/2") # plot
plt.plot(3,6.2,"o",markersize=np.sqrt(50.),color="r") # plot P vs. time
#plt.title('Close Up', fontsize = 8,color='m') # set title
plt.xlabel('t', fontsize = 8)
plt.xticks(fontsize = 8) # adjust the fontsize
plt.yticks(fontsize = 8) # adjust the fontsize
plt.axis([0,10, 0,10]) # set the range of the axes
plt.legend(fontsize=8) # show the legend

plt.show() # necessary for some platforms

# save the figure as .jpg (other formats: png, pdf, svg, (ps, eps))
#fig.savefig('BGTOAGTO.jpg', dpi=fig.dpi, bbox_inches = "tight")
fig.savefig('graph4.jpg', dpi=fig.dpi, bbox_inches = "tight")

```